NARSEE MONJEE INSTITUTE OF MANAGEMENT STUDIES SCHOOL OF BUSINESS MANAGEMENT



Leveraging Machine Learning in Business Applications Individual Assignment (2025-2026)

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Division – D

Topic - Data-Driven Credit Risk Optimization for Retail Lending Institutions

Submitted to: Dr. Rajesh Save

Executive Summary:

Zenith FinBank is a mid-sized retail banking organization that operates in India, providing personal, education, auto, and small business loans. However, over the past few years, the bank has been experiencing a steady increase in the number of loan defaults, resulting in **higher Non-Performing Assets (NPAs) and lower profitabilit**y. The current credit risk assessment process is highly dependent on traditional credit scoring and manual credit assessment, making it difficult for the bank to proactively identify high-risk borrowers.

To mitigate this issue, Zenith FinBank intends to use machine learning approaches to create a predictive model that can predict the probability of a loan default at the time of loan application. By using past loan data with demographic, financial, and behavioral characteristics of the borrowers, the bank intends to enhance credit risk assessment, minimize defaults, and improve overall portfolio quality. The final goal is to enable informed lending decisions that strike a balance between growth and risk management.

Problem Statement:

Can we effectively forecast whether a loan applicant has a high chance of defaulting on the loan based on demographic, financial, and loan-related variables that are available at the point of loan application?

In other words, the bank would like to develop a **binary classification model** that helps to **identify high-risk loan applicants in advance** so that risk can be mitigated proactively.

Business Domain and Context:

Zenith FinBank is engaged in the retail lending business, providing various types of loan products such as personal loans, car loans, education loans, and small business loans. The bank handles a large number of loan applications on a monthly basis and competes fiercely in a growing but cautious financial environment.

Over the past few years, the bank has noticed an increase in the number of loan defaults, which has affected the bank's profitability and resulted in an increase in Non-Performing Assets (NPAs). Loan defaults not only result in a reduction in interest income but also result in increased costs of recovery, legal charges, and capital provisioning. Achieving a balanced approach between loan growth and credit risk management has thus become a key strategic focus.

At present, the bank uses conventional credit evaluation processes that involve credit score checks, income verification, debt-to-income ratio analysis, and manual credit evaluation. Although these processes enable the bank to make a preliminary assessment of the credit risk,

they do not make optimal use of the extensive historical data available. Additionally, manual credit evaluation can be time-consuming, unstandardized, and prone to human error.

By integrating machine learning into the loan approval process, the bank aims to:

- Identify high-risk applicants before loan disbursement
- Improve the accuracy and consistency of credit risk evaluation
- Optimize interest rate pricing based on risk levels
- Reduce default rates while maintaining healthy loan growth

A predictive model would serve as a decision-support tool for credit officers, enabling data-driven lending strategies and improved portfolio performance.

Data Description and Variables:

The dataset consists of 255,347 historical loan records with 18 attributes representing borrower demographics, financial characteristics, employment details, and loan-specific information. The data includes both independent variables (features) and one dependent variable (target).

Link to Python Code:

https://colab.research.google.com/drive/1wDLBcPCsVETS0-DtyejkAguvdUHTOjf5?usp=sharing

Dependent Variable (DV):

Default

- **Type:** Binary (Integer)
- **Values:**

a) 1 = Loan Defaulted
b) 0 = Loan Not Defaulted

This is the target variable the model aims to predict. The objective is to classify applicants into high-risk (likely to default) and low-risk (unlikely to default) categories.

Independent Variable (IV):

A) Demographic Features:

- **Age (Integer) –** Age of the borrower.
- **MaritalStatus (Categorical: Divorced, Married, Single) -** Marital Status of the borrower.
- **HasDependents (Categorical: Yes/No) –** Indicates whether the borrower has financial dependents.
- **Education (Categorical: PhD, Master's, Bachelor's, High School) –** Highest level of education attained.

B) Financial Features:

- **Income (Integer):** Annual Income of the borrower.

- **CreditScore (Integer):** Creditworthiness score assigned by the credit bureau.
- **DTIRatio (Float):** Debt-to-Income ratio indicating debt obligations relative to income.
- **NumCreditLines (Integer):** Number of active credit lines held by the borrower.

C) Employment Features:

- **EmploymentType (Categorical: Full-time, Part-time, Self-employed, Unemployed):** Type of employment status.
- **MonthsEmployed (Integer):** Number of months the borrower has been employed.

D) Loan-Specific Features:

- **LoanAmount (Integer):** Amount of money borrowed.
- **LoanTerm (Integer):** Loan duration in months.
- **InterestRate (Float):** Interest rate applied to the loan.
- **LoanPurpose (Categorical: Home, Auto, Education, Business, Other):** Purpose for which the loan was taken.
- **HasMortgage (Categorical: Yes/No):** Indicates whether the borrower currently has a mortgage.
- **HasCoSigner (Categorical: Yes/No):** Indicates whether the loan has a co-signer.

Excluded Variable:

- **LoanID (Identifier/Nominal):** Unique identifier for each loan record. This variable will not be used for modeling as it does not contribute predictive value.

Data Cleaning:

```
Missing Values per Column:
LoanID             0
Age                0
Income             0
LoanAmount         0
CreditScore        0
MonthsEmployed     0
NumCreditLines     0
InterestRate       0
LoanTerm           0
DTIRatio           0
Education          0
EmploymentType     0
MaritalStatus      0
HasMortgage        0
HasDependents      0
LoanPurpose        0
HasCoSigner        0
Default            0
dtype: int64

Total Missing Values: 0
```

```
LoanID - Empty Strings: 0
Education - Empty Strings: 0
EmploymentType - Empty Strings: 0
MaritalStatus - Empty Strings: 0
HasMortgage - Empty Strings: 0
HasDependents - Empty Strings: 0
LoanPurpose - Empty Strings: 0
HasCoSigner - Empty Strings: 0

Duplicate Rows: 0

Range Checks:
Age Range: 18 - 69
Income Range: 15000 - 149999
LoanAmount Range: 5000 - 249999
CreditScore Range: 300 - 849
DTI Ratio Range: 0.1 - 0.9

Shape after logical cleaning: (255347, 18)

Outlier Capping Completed.

Final Cleaned Shape: (255347, 18)
```

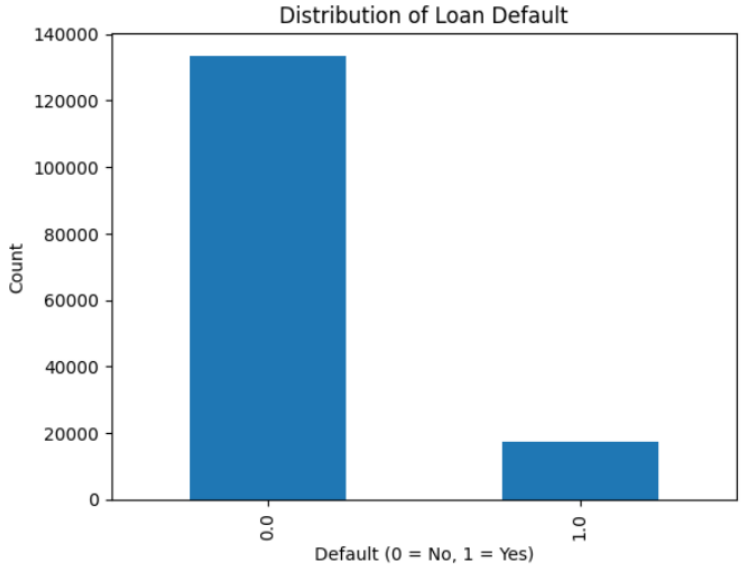
An initial data quality assessment was conducted to evaluate the completeness, consistency, and validity of the dataset. The analysis revealed that all 255,347 loan records are structurally complete, with no missing values, null entries, or duplicate records across any of the 18 variables. Additionally, range validation checks confirmed that all numerical features fall

within realistic and logically acceptable boundaries (e.g., age between 18 and 69 years, credit score between 300 and 849, and DTI ratio between 0.1 and 0.9). No negative or implausible values were observed. Outlier assessment using the IQR method did not significantly alter the dataset, indicating that the data distribution is well-controlled. Overall, the dataset demonstrates high structural integrity and is suitable for advanced exploratory analysis and predictive modeling.

Exploratory Data Analysis:

```
Counts:
 Default
0.0    133532
1.0     17403
Name: count, dtype: int64

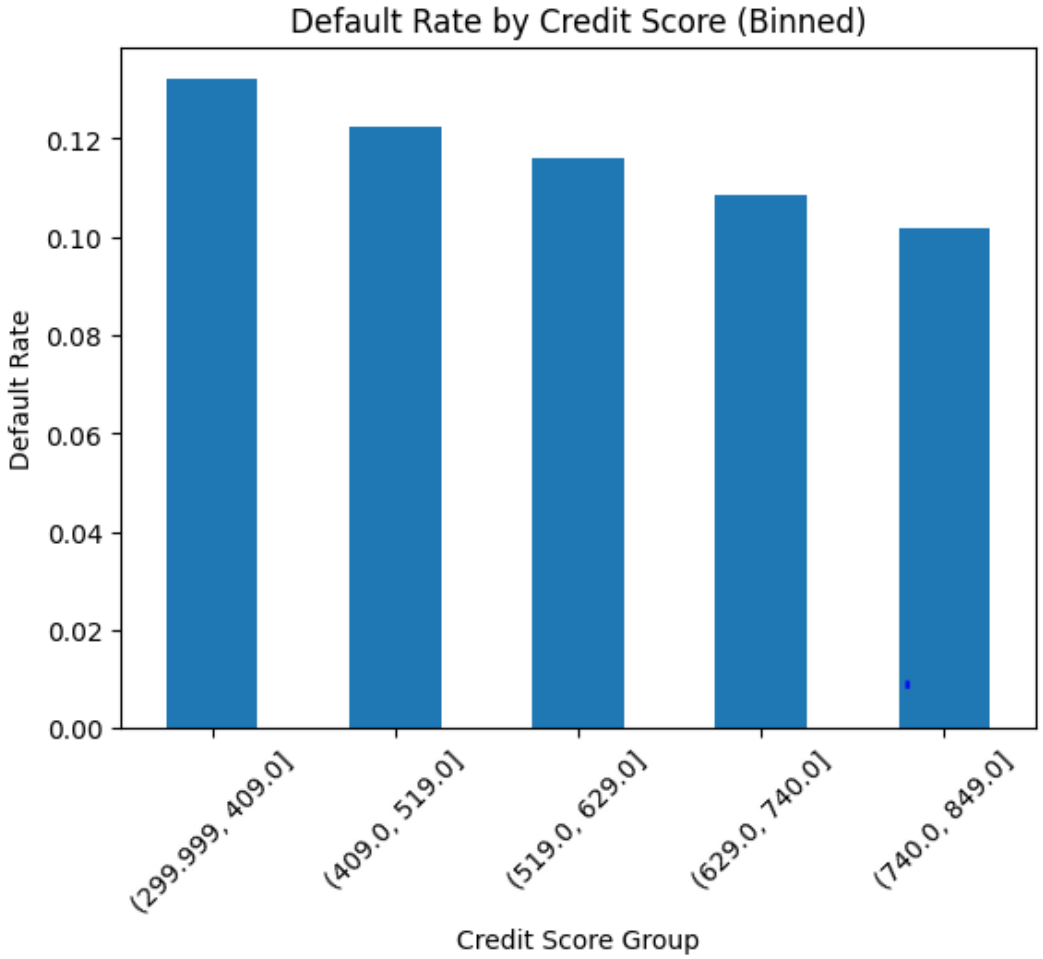
Percentage:
 Default
0.0    88.469871
1.0    11.530129
Name: proportion, dtype: float64
```



The dataset exhibits class imbalance, with approximately 11.53% of loans resulting in default. This imbalance may bias machine learning models toward predicting the majority class (non-default). Therefore, appropriate evaluation metrics such as Precision, Recall, F1-score, and ROC-AUC will be prioritized over accuracy during model evaluation.

Default Rate by Credit Score

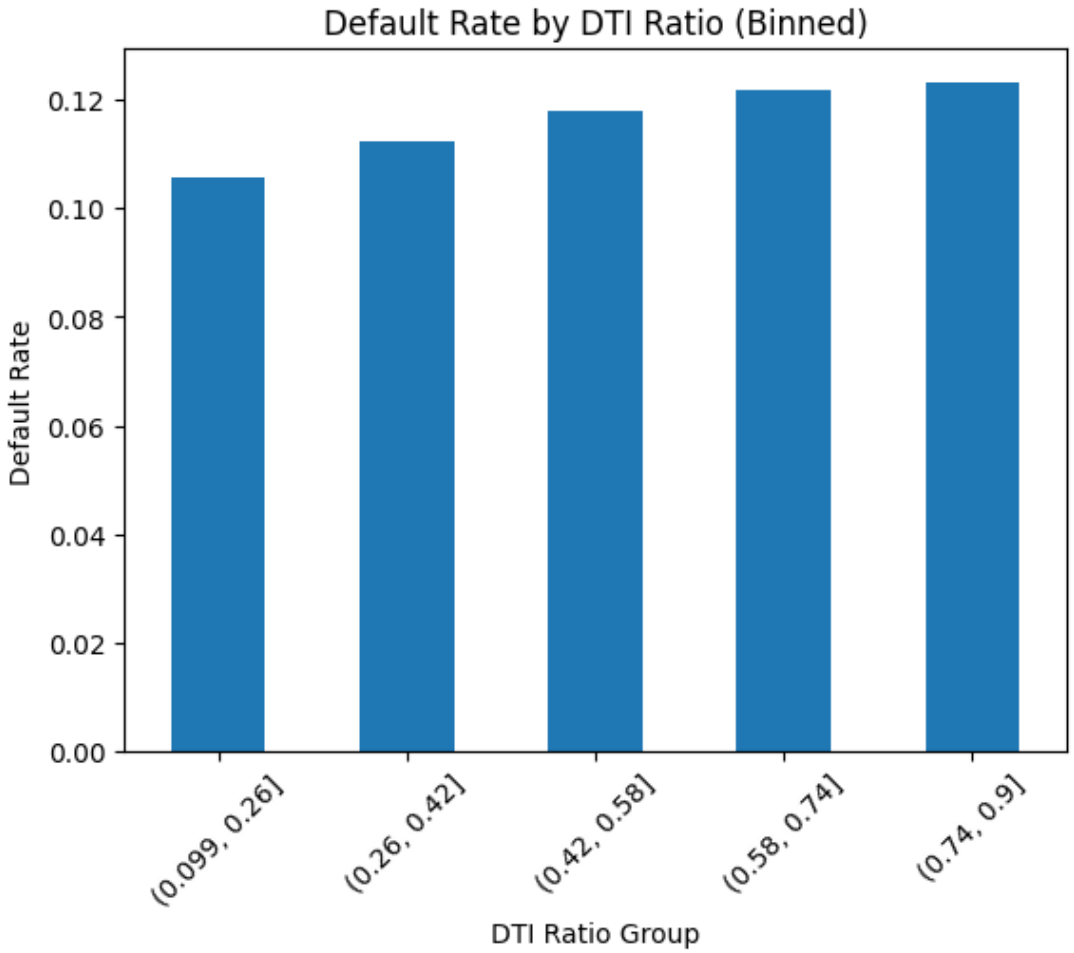
```
Default Rate by Credit Score Group:
CreditScore_Bin
(299.999, 409.0]    0.132070
(409.0, 519.0]      0.122278
(519.0, 629.0]      0.115975
(629.0, 740.0]      0.108462
(740.0, 849.0]      0.101672
Name: Default, dtype: float64
```



The analysis of default rates across credit score segments reveals a clear inverse relationship between creditworthiness and loan default probability. Borrowers in the lowest credit score bracket (300–409) exhibit the highest default rate at approximately 13.2%, while those in the highest bracket (740–849) show a lower default rate of around 10.2%. The steady decline in default rates across increasing credit score groups indicates a monotonic negative relationship, suggesting that credit score is a meaningful predictor of credit risk. However, the relatively moderate spread between the highest and lowest risk groups implies that credit score alone may not fully explain default behavior, highlighting the importance of incorporating additional financial variables into the predictive model.

Default Rate by DTI Ratio

```
Default Rate by DTI Ratio Group:
DTI_Bin
(0.099, 0.26]    0.105785
(0.26, 0.42]     0.112417
(0.42, 0.58]     0.117950
(0.58, 0.74]     0.121884
(0.74, 0.9]      0.123101
Name: Default, dtype: float64
```

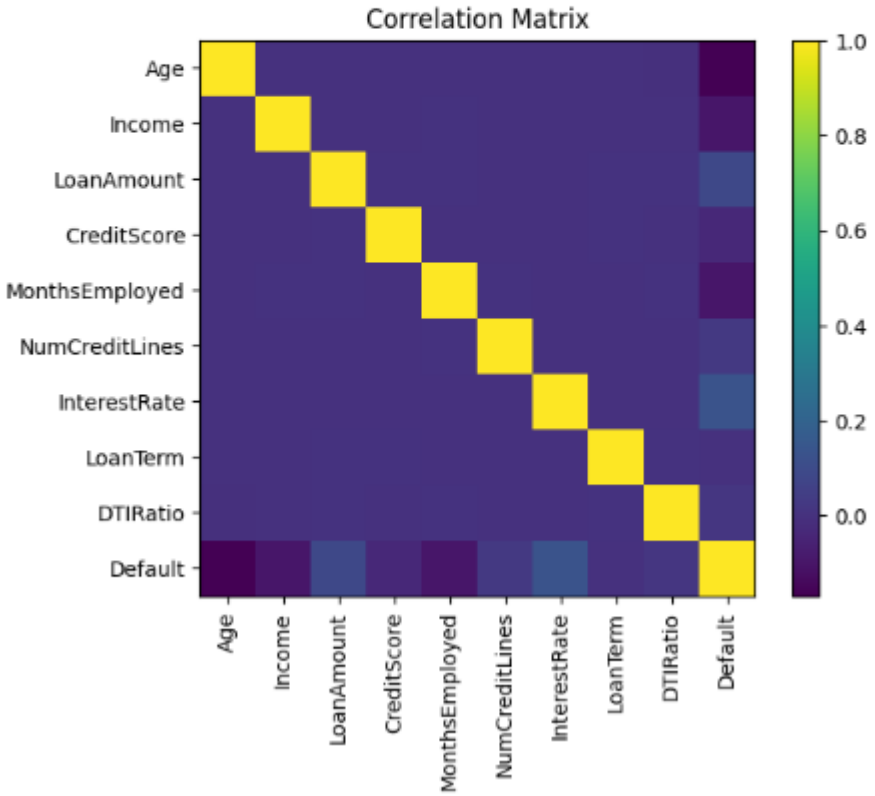


The analysis of default rates across Debt-to-Income (DTI) ratio segments reveals a clear positive relationship between financial leverage and loan default probability. Borrowers with lower DTI ratios (0.10–0.26) exhibit a default rate of approximately 10.6%, whereas those in the highest DTI bracket (0.74–0.90) show an increased default rate of approximately 12.3%. The steady upward trend suggests that higher debt burdens relative to income significantly increase repayment risk. This monotonic pattern indicates that DTI ratio is a meaningful financial stress indicator and is likely to serve as an important predictor in the credit risk modeling framework.

Multicollinearity:

```
Correlation with Default:

Default           1.000000
InterestRate      0.131273
LoanAmount        0.086659
NumCreditLines    0.028330
DTIRatio          0.019236
LoanTerm          0.000545
CreditScore      -0.034166
MonthsEmployed   -0.097374
Income           -0.099119
Age              -0.167783
Name: Default, dtype: float64
```

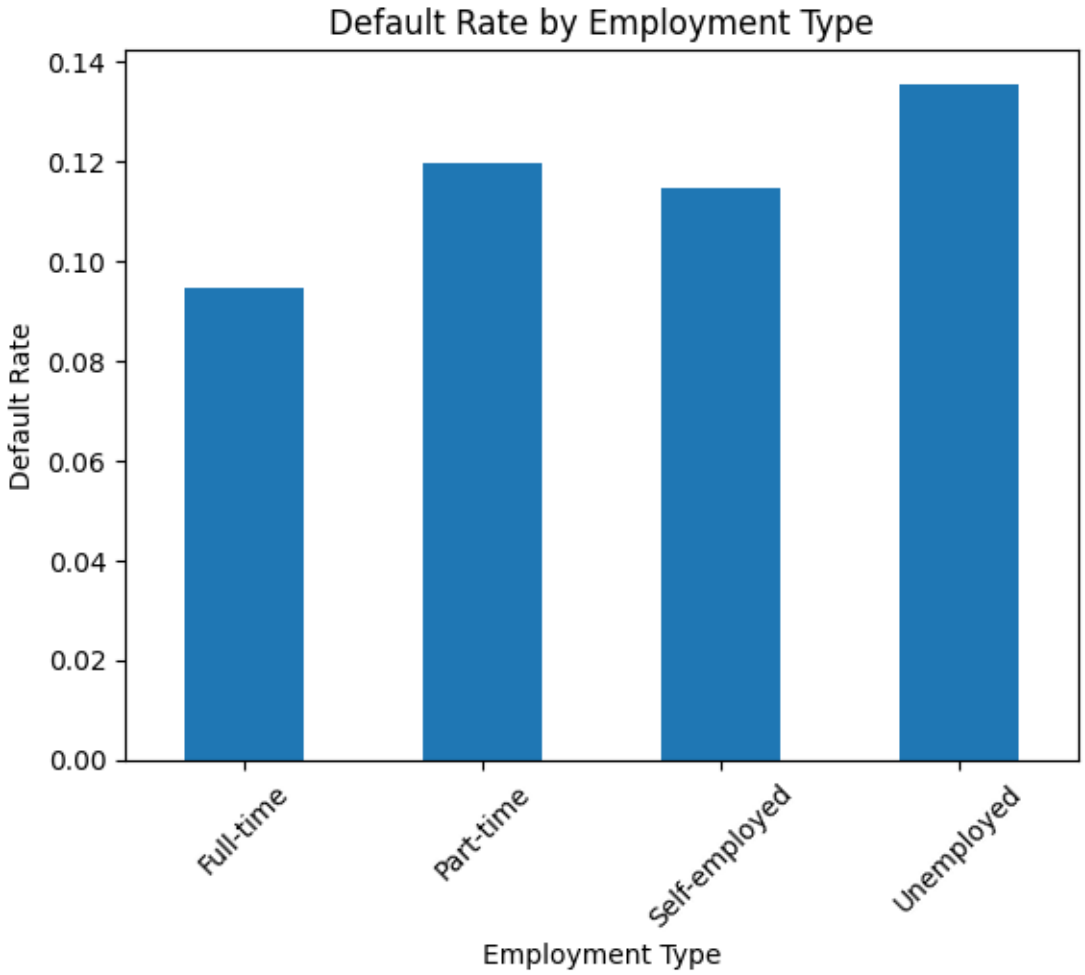


The correlation analysis indicates that no single variable exhibits a strong linear relationship with loan default, suggesting that default behavior is influenced by multiple interacting factors. Age shows the strongest negative correlation (-0.168), implying that older borrowers are less likely to default. Interest rate demonstrates the highest positive correlation (0.131), indicating that loans with higher interest rates are associated with increased default risk. Income and employment stability variables also display moderate negative correlations, reinforcing the importance of financial capacity in credit risk assessment. The relatively weak correlations across individual predictors highlight the need for multivariate modeling approaches to accurately capture default risk.

Default Rate by Employment Type:

```
Default Rate by Employment Type:

EmploymentType
Full-time        0.094634
Part-time        0.119652
Self-employed    0.114620
Unemployed       0.135529
Name: Default, dtype: float64
```

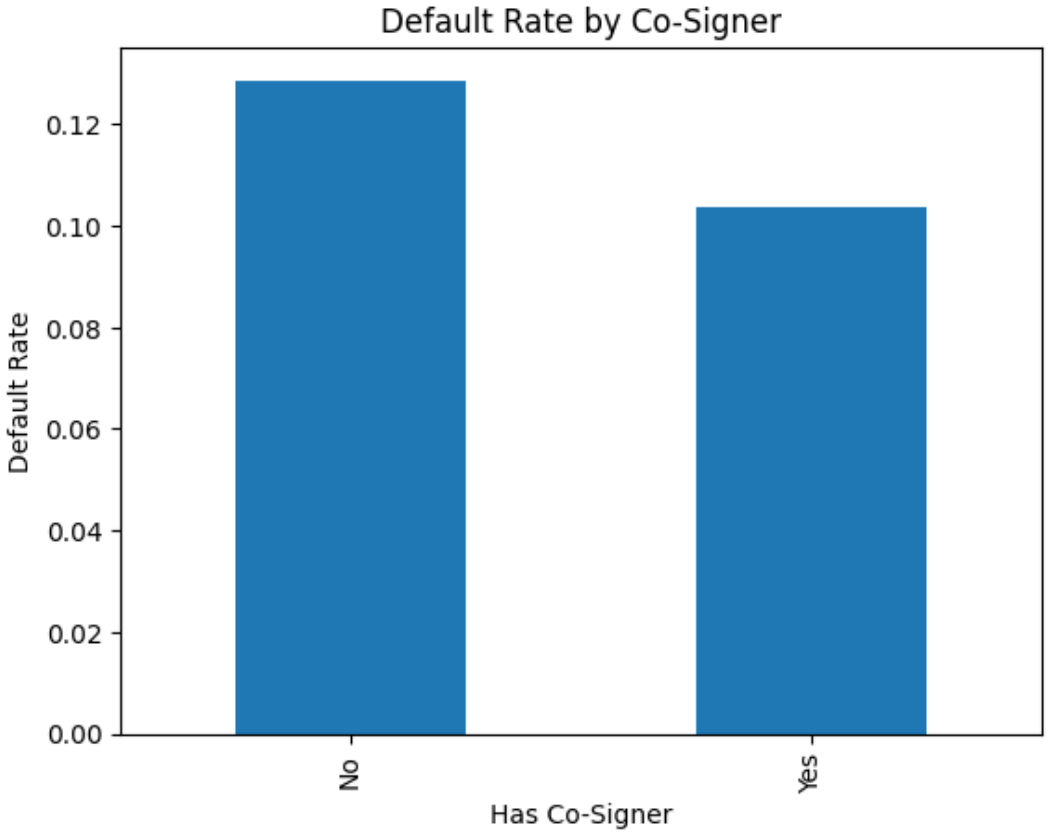


The analysis of loan default rates across employment categories reveals a clear relationship between employment stability and credit risk. Unemployed borrowers exhibit the highest default rate at approximately 13.6%, followed by part-time and self-employed individuals. In contrast, full-time employed borrowers demonstrate the lowest default rate at 9.5%, indicating greater financial stability and repayment capacity. The approximately 4 percentage point spread between the lowest and highest risk groups highlights employment status as a significant categorical predictor of loan default. This suggests that employment stability plays an important role in credit risk assessment and should be incorporated into the predictive modeling framework.

Default Rate by Co-Signer

```
Default Rate by Co-Signer:

HasCoSigner
No     0.128661
Yes    0.103601
Name: Default, dtype: float64
```



The analysis of default rates based on co-signer status indicates that loans backed by a co-signer exhibit lower default probability (10.36%) compared to loans without a co-signer (12.87%). The approximately 2.5 percentage point reduction in default risk suggests that co-signers provide additional financial assurance and reduce repayment uncertainty. This finding highlights co-signer presence as a meaningful categorical predictor of credit risk and presents a potential policy lever for mitigating default rates among higher-risk applicants.

Logistic Regression Output:

```
Accuracy: 0.6765028392402584
Precision: 0.21963359101980304
Recall: 0.6993761591637161
F1 Score: 0.3342869806987146
ROC-AUC: 0.7531790420170843

Confusion Matrix:
[[30401 14738]
 [ 1783  4148]]
```

The baseline logistic regression model demonstrates strong discriminatory power with an ROC-AUC score of 0.753, indicating good ability to distinguish between defaulters and non-defaulters. The model achieves a recall of approximately 70%, successfully identifying a significant proportion of high-risk borrowers. However, the precision remains relatively low (21.9%), suggesting a high number of false positives. This trade-off is expected due to class imbalance and the use of balanced class weights, which prioritize risk detection over overall accuracy. While the model effectively captures default risk, further refinement may improve precision and reduce false alarms.

Metrics of Logistic Regression at different thresholds:

Threshold	Precision	Recall	F1
0.1	0.119	0.995	0.213
0.2	0.133	0.972	0.233
0.3	0.153	0.915	0.262
0.4	0.181	0.825	0.297
0.5	0.220	0.699	0.334
0.6	0.272	0.546	0.363
0.7	0.340	0.352	0.346
0.8	0.454	0.156	0.232

Threshold analysis indicates that the default 0.5 probability cutoff does not yield the optimal precision-recall balance. The highest F1 score (0.363) is achieved at a threshold of 0.6, suggesting that increasing the classification cutoff improves overall model balance. At this threshold, the model achieves improved precision while maintaining moderate recall, resulting in a more stable trade-off between identifying defaulters and minimizing false positives. This demonstrates the importance of threshold tuning in imbalanced classification problems, particularly in credit risk modeling where business objectives influence decision boundaries.

Random Forrest Output:

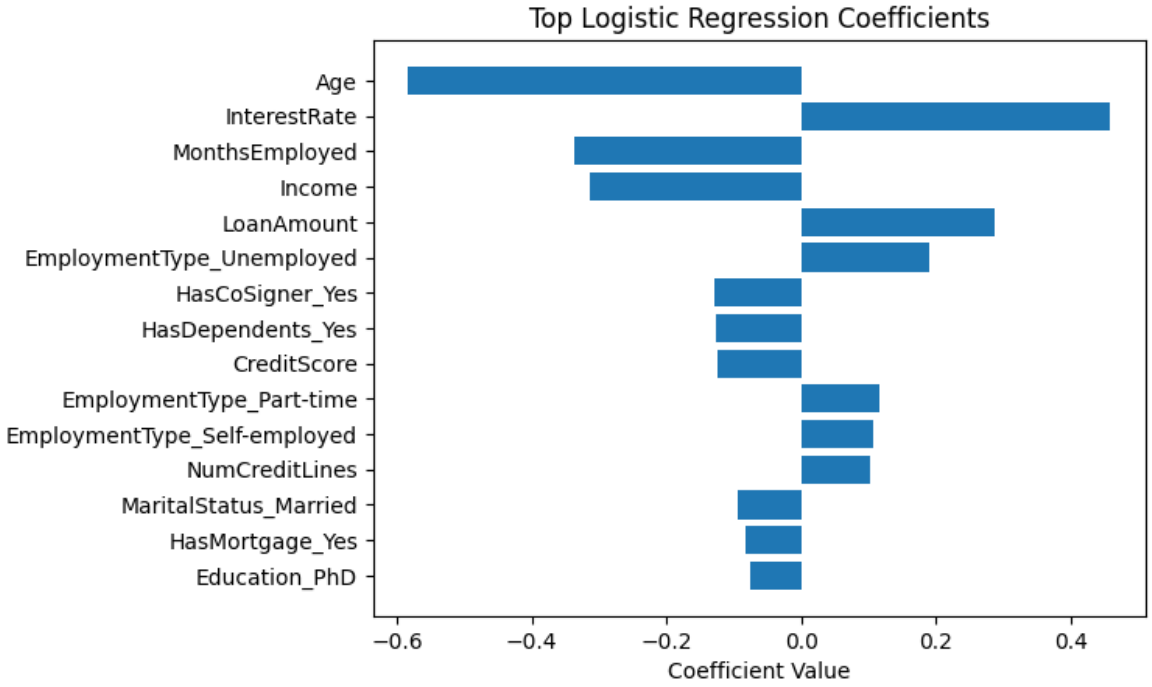
```
Accuracy: 0.884726845506168
Precision: 0.7244897959183674
Recall: 0.011970999831394368
F1 Score: 0.02355282799800962
ROC-AUC: 0.7436506349825389

Confusion Matrix:
[[45112    27]
 [ 5860    71]]
```

While the Random Forest model achieved higher overall accuracy (88.5%) and precision, it exhibited extremely low recall (1.2%), indicating poor detection of defaulters. In contrast, the Logistic Regression model achieved a significantly higher recall (69.9%) and a slightly superior ROC-AUC score (0.753), demonstrating better discrimination ability. Given that identifying high-risk borrowers is critical in credit risk management, the Logistic Regression model provides a more appropriate balance between detection capability and overall performance. Therefore, Logistic Regression is selected as the preferred modeling approach.

Feature Importance:

	Feature	Coefficient
0	Age	-0.583092
6	InterestRate	0.458738
4	MonthsEmployed	-0.337433
1	Income	-0.313960
2	LoanAmount	0.287877
14	EmploymentType_Unemployed	0.190031
23	HasCoSigner_Yes	-0.129242
18	HasDependents_Yes	-0.126034
3	CreditScore	-0.125004
12	EmploymentType_Part-time	0.116864
13	EmploymentType_Self-employed	0.107270
5	NumCreditLines	0.102526
15	MaritalStatus_Married	-0.093929
17	HasMortgage_Yes	-0.082065
11	Education_PhD	-0.075899



The logistic regression coefficients reveal that age and interest rate are the strongest predictors of loan default. Age exhibits the largest negative coefficient, indicating that older borrowers are significantly less likely to default. Conversely, higher interest rates substantially increase default probability, reflecting both increased repayment burden and risk-based pricing mechanisms. Employment stability variables, including months employed and employment type, also demonstrate strong influence, confirming that income consistency plays a critical role in credit risk. Additionally, higher income, presence of a co-signer, and stronger credit scores reduce default likelihood. Overall, the model highlights the multi-dimensional nature of credit risk, where demographic stability, financial capacity, and loan structure jointly determine repayment outcomes.

Success Metric:

A successful deployment of the loan default prediction model should be judged not only on its statistical performance but also on its business impact. In terms of statistical performance, the first success criterion is the ROC-AUC score, which measures the model's ability to distinguish between defaulters and non-defaulters. The Logistic Regression model has an ROC-AUC score of 0.753, which is excellent. Secondly, recall is given more importance over accuracy because of the class imbalance problem, as it is very important to identify the high-risk borrowers in order to avoid maximum financial losses. The optimal threshold of 0.6 offers the best trade-off between precision and recall, as evidenced by the highest F1-score.

In terms of financial impact, a better prediction of defaults would help the bank decrease Non-Performing Assets (NPAs), lower recovery and legal costs, and make better use of capital. A small decrease in the default rate can greatly enhance the profitability of the portfolio because of the sheer volume of retail lending.

From an operational perspective, the integration of the predictive model within the loan approval system could potentially decrease the manual underwriting process, increase the consistency of decisions, and improve the speed of loan approvals. The predictive model could be used as a risk scoring component to assist credit managers in making decisions.

From a customer service perspective, risk-based pricing made possible through predictive modeling could potentially allow low-risk customers to enjoy improved interest rates, thus increasing customer satisfaction and remaining profitable.

In general, success would be measured by a combination of factors including good model discrimination, the ability to identify high-risk customers, improved financial performance, and increased operational efficiency.

Potential Challenges:

However, despite its robust performance, there are a number of issues that need to be taken into consideration when implementing the loan default prediction model in a practical banking setting.

1) **Data Limitations:**

While the data used in this study is clean, in practical banking settings, there could be issues of missing data, errors in reporting, time lags in updates, and inconsistencies. Furthermore, the data may not include macroeconomic factors such as inflation rates, unemployment trends, and economic shocks, which could have a material impact on default behavior.

2) **Class Imbalance:**

With only 11.53% of the data representing loans that have defaulted, the data is moderately imbalanced. This could pose a problem in striking a balance between recall and precision. There could be a tendency to overemphasize recall, which could result in an unacceptably high number of false positives, potentially denying loans to deserving customers.

3) **Model Bias and Fairness:**

There could be fairness and ethical issues related to some demographic variables (such as age or employment status). In banking, there are regulatory requirements for explainability and fairness in credit decisions.

4) **Interpretability and Regulatory Requirements:**

Financial organizations are strictly regulated. The models employed for credit risk analysis need to be interpretable and explainable. Although Logistic Regression is quite interpretable, there is a need for periodic validation and documentation to meet the regulatory requirements.

5) **Deployment and Monitoring:**

The model performance could deteriorate over time because of changes in customer behavior or economic trends (concept drift). There is a need for periodic monitoring,

retraining, and recalibration of the decision boundary to ensure that the model continues to perform well.

Conclusion:

In conclusion, although the predictive model has immense potential to improve credit risk analysis, its successful implementation needs proper governance, monitoring, and adherence to regulatory requirements.